

UCT513: MACHINE LEARNING

L	T	P	Cr
2	1	2	3.5

Course Objectives: This course provides a broad introduction to machine learning and statistical pattern recognition. It offers some of the most cost-effective approaches to automated knowledge acquisition in emerging data-rich disciplines and focuses on the theoretical understanding of these methods, as well as their computational implications.

Introduction to Machine Learning (ML); Relationship between ML and human learning, A quick survey of major models of how machines learn, Example applications of ML

Classification: Supervised Learning; The problem of classification, Feature engineering, Training and testing classifier models, Cross-validation, Model evaluation (precision, recall, F1-measure, accuracy, area under curve), Statistical decision theory including discriminant functions and decision surfaces, Naive Bayes classification, Bayesian networks, Decision Tree and Random Forests, k-Nearest neighbor classification, Support Vector Machines, Artificial neural networks including backpropagation, Applications of classifications, Ensembles of classifiers including bagging and boosting.

Hidden Markov Models (HMM); HMM with forward-backward and Viterbi algorithms, Sequence classification using HMM, Conditional random fields, Applications of sequence classification such as part-of-speech tagging.

Regression: Multi-variable regression; Model evaluation, Least squares regression, Regularization, LASSO, Applications of regression.

Association rule mining: Association rule mining algorithms including apriori

Expectation-Maximization: EM algorithm for unsupervised learning, Clustering: average linkage; Ward's algorithm, Minimum spanning tree clustering, K-nearest neighbors clustering, BIRCH, CURE, DBSCAN

Anomaly and outlier detection methods

Laboratory Work:

Implement data preprocessing, Simple Linear Regression, Multiple Linear Regression, Decision Tree, Random forest classification, Naïve Bayes algorithm; K-Nearest Neighbors (K-NN), Support Vector Machine, k-Means, Apriori algorithm in Python.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Analyze methods and theories in the field of machine learning and provide an introduction to the basic principles, techniques, and applications of machine learning, supervised, unsupervised and reinforcement learning.
2. Comprehend and apply different classification and clustering techniques.
3. Comprehend and apply different association mining techniques.
4. Apply the algorithms to a real problem & optimize the models learned.

Text Books:

1. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, Machine Learning with Python, BPB Publication (2018).

Reference Books:

1. Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Edition.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994).